

[Dashboard](#) / [Meine Kurse](#) / [2023S510105](#) / [28.06.2023 \(Written Exam\)](#) / [START EXAM \(28.06.2023\)](#)

Begonnen am Wednesday, 28. June 2023, 10:26

Status Beendet

Beendet am Wednesday, 28. June 2023, 11:08

Verbrauchte Zeit 42 Minuten 42 Sekunden

Bewertung Bisher nicht bewertet

Frage 1

Vollständig

Erreichbare Punkte: 4,00

List **2 core findings** from each of the **scientific papers** we read on "Calibrated Trust in AI" (Leichtmann et al., 2023) and "Human-AI Creativity" (Inie et al., 2023).

Would you classify each of the works as **quantitative or qualitative** research? Briefly justify your answer.

Calibrated trust in AI:

1. XAI methods have a significant positive effect on participants decision performance
2. XAI and lack of trust into AI system were found to be associated

I would say that the research paper can be classified as qualitative research, as a phenomenon that is not well-studied (human and AI relationship) was studied. Additionally, new theories or hypothesis were developed.

Human-AI Creativity:

1. Study authors were able to categorize common concerns and positive expectations that artists/creatives have in regards to AI
2. Study authors found that creatives have a superficial understanding of AI

I would also classify this paper as a qualitative research, for the same two reasons I outlined as above: AI is a new technology, so the phenomenon (creatives' reactions and opinions on generative AI) were studied.

Frage 2

Vollständig

Erreichbare Punkte: 4,00

What is meant by the **replicability crisis**?

Describe **3 of its causes**, as discussed among researchers.

In early 2010s (I believe 2011, but not sure) it was found that a famous researcher has faked a lot of data that was used in many of his research papers. This led to a thorough investigation of his works, and this investigation continued in works of other people. It was found that it was systematically impossible to reproduce the results of the studies. The problem was then also found in other research areas (not only psychology). There are many reasons for this, some of which are:

1. Bad/questionable research practices - sample sizes of people interviewed, where those people came from (their background - we had an example of "99.98% of people who took our survey like taking surveys"), etc.
2. Problems in quality of reporting/reporting standards - bad documentation of how a study was done, missing data (statistics, research process, study details). In general: opaque (non-transparent) research practices
3. Publication bias - articles with big/interesting results are more likely to be published. If you don't find anything, you might not write a paper/it might not get published.

Frage 3

Vollständig

Erreichbare Punkte: 4,00

Briefly describe the **3 Basic Psychological Needs** and name the **theory** from which they are derived. Discuss on the example of one of these needs, how using AI may **threaten** it.

The theory describing the three basic psychological needs is the **Self-Determination Theory**. In order to be self-determined, all three needs to be satisfied. The three basic needs are:

1. **Competence**: the experience of being good at something, of mastering it. Satisfaction of this leads to a feeling of, well, satisfaction, allowing us to improve our skills even further, to become more skilled and effective. The opposite of it (failing to achieve something) leads to frustration, which leads to a feeling of ineffectiveness and helplessness.
2. **Autonomy**: the experience of being "free", of willingness. Satisfaction of this basic need leads to a feeling of integrity. Failure to satisfy it leads to a feeling of pressure and stress.
3. **Relatedness**: the experience of belonging. Satisfaction leads to a feeling of inclusion and belonging. Failure to satisfy leads to a feeling of loneliness and social alienation.

Using an AI system to automate a process may rob the human of a feeling of competence. If an AI starts helping you in your job, it may make your life easier. But as the AI starts helping you more and more, and as the AI gets more competent and can do more things, it may be doing the job *instead* of you. Then, the human might start to feel dissatisfied, as they have a feeling that they are not competent anymore, or not needed anymore.

Another example: AI telling the person to do something might rob them of the feeling of autonomy. The human may feel like a slave to machine's commands.

Frage 4

Vollständig

Erreichte Punkte 1,00 von 1,00

What is a good example of a so-called **jangle fallacy** in empirical research?

- ☐ In an experiment, it turns out that the discriminant validity of the independent variable and the dependent variable is not sufficient, but the results are published anyway.
- ☐ Three research papers claim that the emotional intelligence of study participants was measured, although the tasks involved were completely different in the studies.
- ☐ Researchers at a university have an app they developed evaluated in a study and recruit their own colleagues and students as participants for the study.
- ☒ Four different researchers all measure how well persons can read emotional states from facial expressions, but the according variable is named differently in each of their studies.

Frage 5

Vollständig

Erreichte Punkte 1,00 von 1,00

What is **not** an assumed consequence of anthropomorphism?

- ☐ Social influence of artificial agents
- ☐ Granting machines the right to moral care
- ☐ Perceiving machines as responsible
- ☒ Dehumanizing real humans

Frage 6

Vollständig

Erreichte Punkte 1,00 von 1,00

Which variable is essential for both **technology acceptance** (according to the TAM) and the assessment of **creativity** (according to typical 2-criterion definitions)?

- ☐ Usability
- ☐ Aesthetic appeal
- ☐ Originality
- ☒ Usefulness

Frage 7

Vollständig

Erreichte Punkte 1,00 von 1,00

Which research design is used in Psychology and other scientific fields to investigate cause-effect relationships (e.g., effects of different light signals on the understandability of a robot)?

- ☐ Foundational review
- ☐ Correlational study
- ☐ Cross-sectional survey
- ☒ Randomized experiment

Frage 8

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about trust is FALSE?

- ☐ Once affective trust is established, a trustor-trustee relationship may become less transparent for objective risk assessments.
- ☐ If an AI system appears competent and reliable, important requirements for the establishment of cognitive trust are met.
- ☒ At the market launch of an AI system, when there is still little practical experience with it, no individual differences in how much users trust the system are expected.
- ☐ Without being vulnerable or at risk, the concept of trust is rather irrelevant.

Frage 9

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the main proposition of the Uncanny Valley hypothesis (Mori, 1970)?

- ☐ Machines should strive for perfect replication of human appearance to avoid any negative response by users.
- ☐ Discomfort experienced by observers has no relation to a machine's behavior or motion.
- ☐ Humans find robots and animated characters unsettling.
- ☒ Machines become unsettling as they approach, but do not perfectly replicate, human appearance and behavior.

Frage 10

Vollständig

Erreichte Punkte 1,00 von 1,00

Everyday psychology refers to non-scientific observations and common sense knowledge about human mind and behavior that people acquire through their daily experiences. What is NOT a valid criticism of everyday psychology?

- ☐ Flawed methods
- ☐ Ad hoc explanations
- ☐ Imprecise language
- ☒ Lack of diversity

Frage 11

Vollständig

Erreichte Punkte 1,00 von 1,00

What does representativeness in empirical research refer to?

- ☐ How well findings obtained under laboratory conditions translate to real-world applications.
- ☐ The statistical significance of the research findings.
- ☐ The process of randomly selecting participants from a larger population.
- ☒ How well a sample reflects characteristics of the larger population from which it is drawn.

Frage 12

Vollständig

Erreichte Punkte 1,00 von 1,00

In the context of human-AI relationships, "calibrated trust" refers to trusting an AI system ...

- ☐ ... based on its sophisticated user interface.
- ☒ ... after ensuring its performance aligns with established benchmarks.
- ☐ ... without any validation or verification.
- ☐ ... based on a generally high personal propensity to trust others.

Frage 13

Vollständig

Erreichte Punkte 0,00 von 1,00

Eudaimonic happiness is often associated with ...

- ☐ ... material possessions and wealth.
- ☐ ... a sense of purpose and meaning.
- ☒ ... external validation and praise.
- ☐ ... pleasure and immediate gratification.

Frage 14

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following statements is FALSE?

The **Alternative Uses Test** ...

- ☐ ... asks participants to come up with unconventional uses for common objects.
- ☐ ... measures convergent thinking capabilities for creative problem-solving.
- ☐ ... is a classic test that has been used for several decades.
- ☒ ... focuses on the fluency, originality, flexibility, and elaboration of participants' responses.

Frage 15

Vollständig

Erreichte Punkte 1,00 von 1,00

Creativity that solves everyday problems is also called

- ☐ Wide i
- ☐ Narrow i
- ☐ Micro c
- ☒ Little c

Direkt zu: